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Department of Artificial Intelligence and Data Science

V Semester B. Tech AIDS

AD23532 – Principles of Data Science

Academic Year: 2026-2027 **Batch:** 2024-2028

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Ex 8: Interactive Dashboards with Tableau

Register Number: 241801238 **Date:** 02-10-2026

Dataset: Sample – Superstore **Tools:** Tableau Public (web authoring)

Aim

To design and implement an interactive dashboard in Tableau using the Sample – Superstore dataset, containing a bar chart of sales by category, a line chart of the sales trend over time and a scatter plot of profit vs sales by sub-category, together with interactive filters and dashboard actions.

Software and Dataset

- **Software:** Tableau Public (web authoring in the browser at public.tableau.com)
- **Dataset:** Sample – Superstore. Tableau Public provides this dataset as the ready-made *Superstore* data source (file *sample_-_superstore.xls*). It has one row per order line, with order and ship dates, customer, segment, region, category, sub-category, sales, quantity, discount and profit. In this version of the dataset the order dates run from 2023 to 2026, and the total sales are $2,326,534.

Task 1: Open Tableau and Load the Dataset

1. Signed in to Tableau Public and clicked **Create a Viz** on the profile page to open a new workbook.
2. In the **Connect to Data** window, selected the **Superstore** data source and clicked **Connect**.

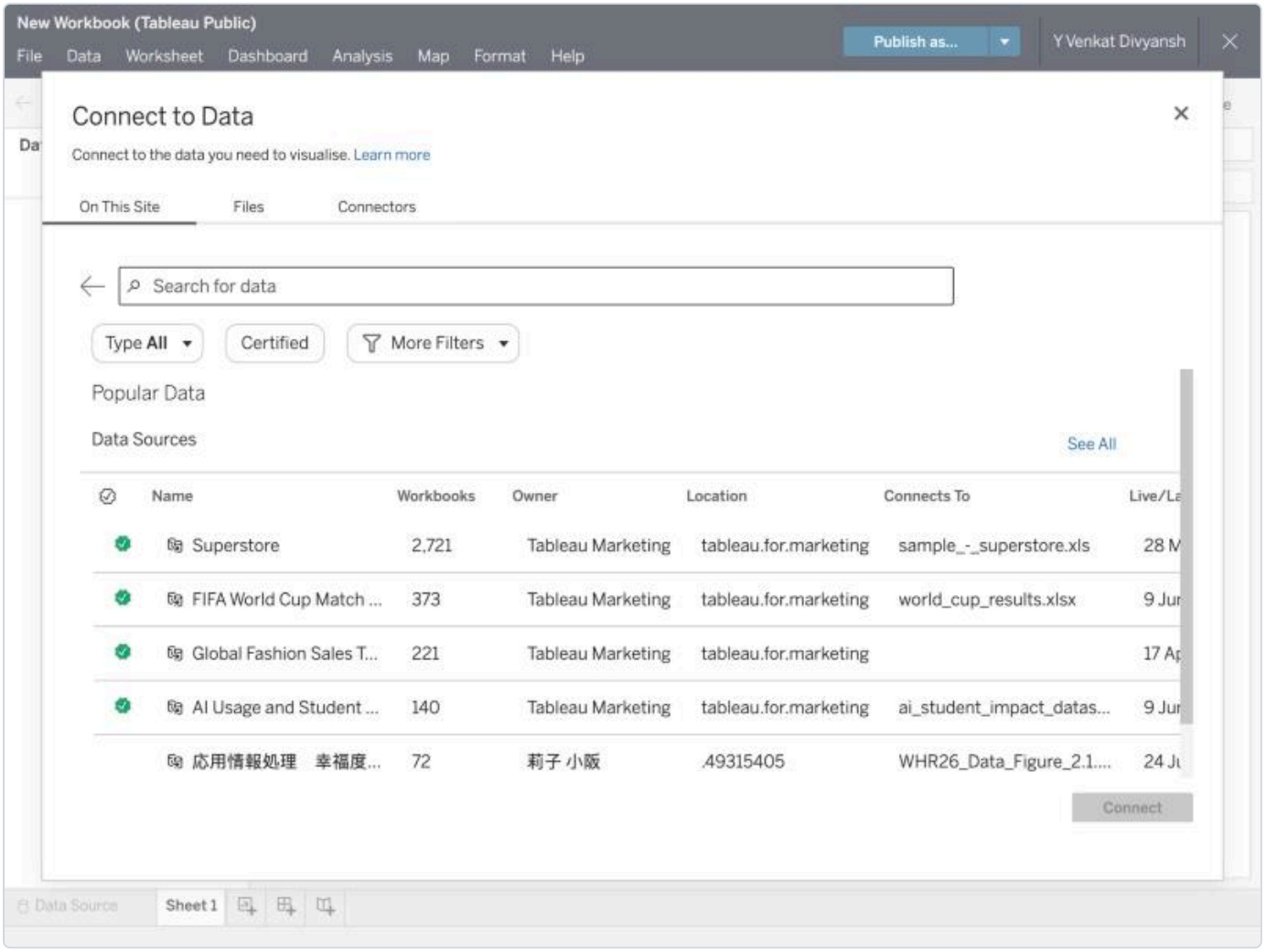


Figure 1: Connect to Data window in Tableau Public with the Superstore data source

Task 2: Explore the Data

After connecting, the **Data** pane lists the fields of the **Orders** table. Tableau splits them into **Dimensions** (categorical fields used to group the data) and **Measures** (numeric fields that are aggregated, for example SUM(Sales)).

Role	Fields
Dimensions	Category, City, Country/Region, Customer ID, Customer Name, Order Date, Order ID, Postal Code, Product ID, Product Name, Region, Row ID, Segment, Ship Date, Ship Mode, State/Province, Sub-Category
Measures	Discount, Profit, Quantity, Sales, Orders (Count)

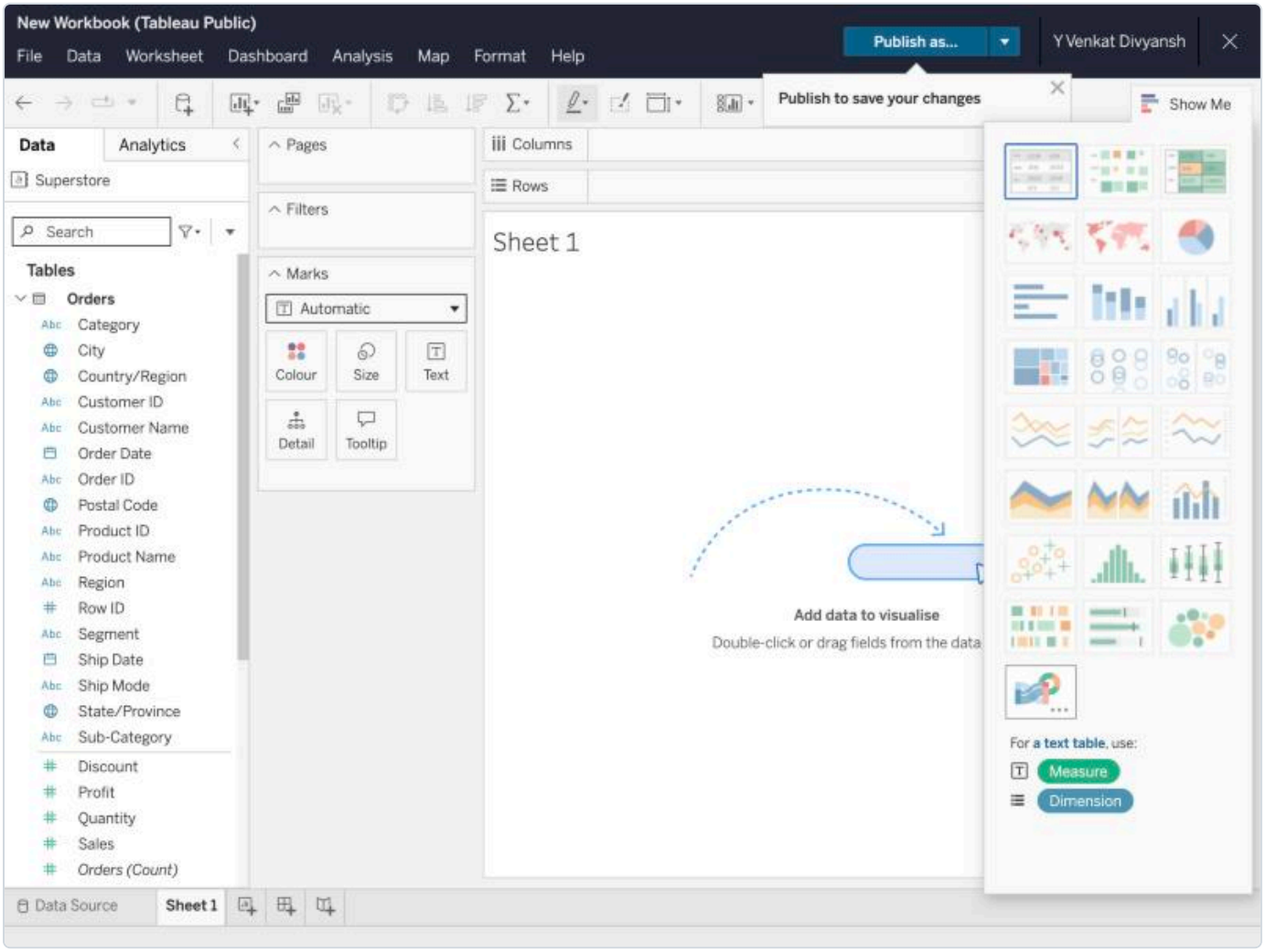


Figure 2: Data pane showing the dimensions (top) and measures (bottom) of the Orders table

Task 3: Worksheet 1 – Bar Chart (Sales by Category)

1. Renamed **Sheet 1** to **Bar_Sales_by_Category**.
2. Dragged **Category** to **Columns** and **Sales** to **Rows** (SUM(Sales)).
3. Clicked **Sort Descending** on the toolbar.
4. Dragged **Region** to **Colour** on the Marks card.
5. Turned on **Show Mark Labels**.
6. Dragged **Region** to **Filters**, kept all values selected, and chose **Show Filter**.

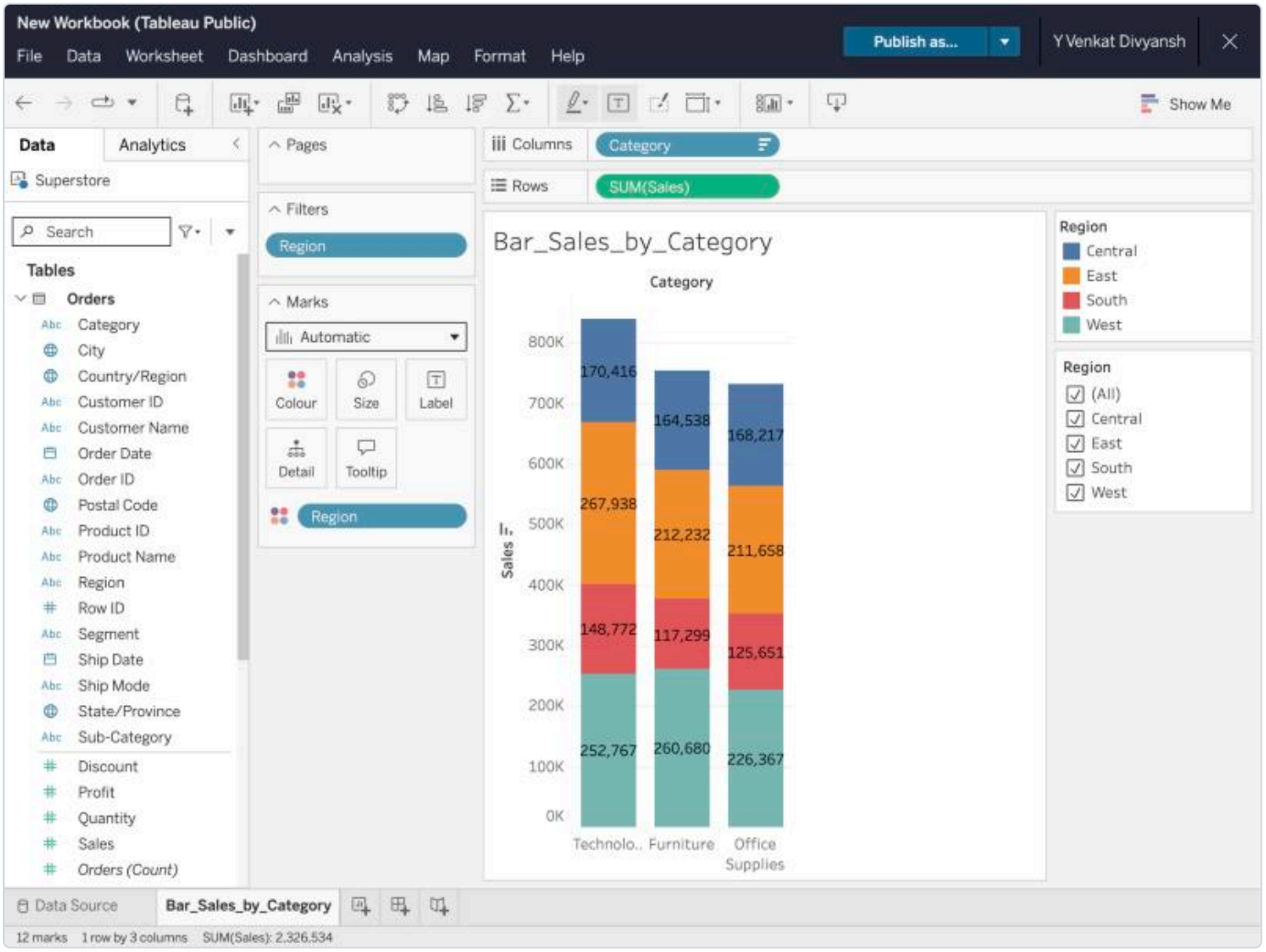


Figure 3: Worksheet 1 – Bar_Sales_by_Category

> **Observation:** Technology has the highest sales ($839,893), followed by Furniture ($754,749) and Office Supplies ($731,893). In every category the West and East regions sell the most and the South sells the least; for example, Technology sales are $267,938 in the East but only $148,772 in the South.

Task 4: Worksheet 2 – Line Chart (Sales Trend Over Time)

1. Added a new worksheet and renamed it **Line_Sales_Trend**.
2. Dragged **Order Date** to **Columns**, right-clicked it and chose **Month** (continuous).
3. Dragged **Sales** to **Rows**.
4. Dragged **Region** to **Colour** to draw one line per region.
5. Formatted the Sales axis as **Currency (Standard)** with the English (United States) locale.
6. Added **Category** as a filter and chose **Show Filter**.

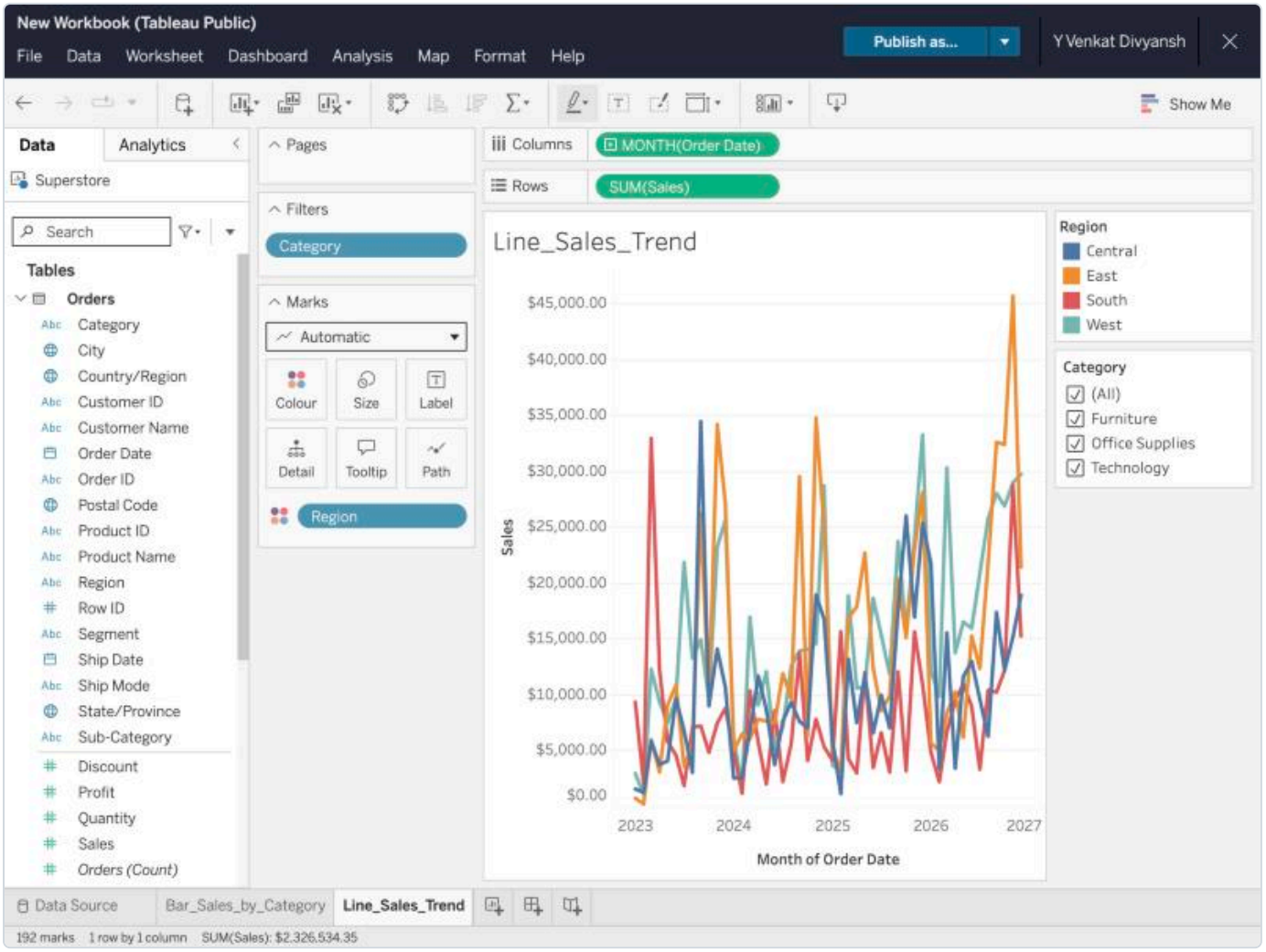


Figure 4: Worksheet 2 – Line_Sales_Trend

> **Observation:** Sales dipped slightly in 2024 and then grew strongly through 2025 and 2026. Each year follows a seasonal pattern, with peaks near the end of the year. The East and West regions reach the highest monthly sales; the highest point is East sales of about $45,000 near the end of 2026. The South region has the lowest sales most of the time.

Task 5: Worksheet 3 – Scatter Plot (Profit vs Sales)

1. Added a new worksheet and renamed it **Scatter_Profit_vs_Sales**.
2. Dragged **Sales** to **Columns** and **Profit** to **Rows**.
3. Dragged **Sub-Category** to **Detail**, so each point is one sub-category (17 points).
4. Dragged **Category** to **Colour**.
5. From the **Analytics** pane, added an **Average Line** for the table, which draws the average on both axes.

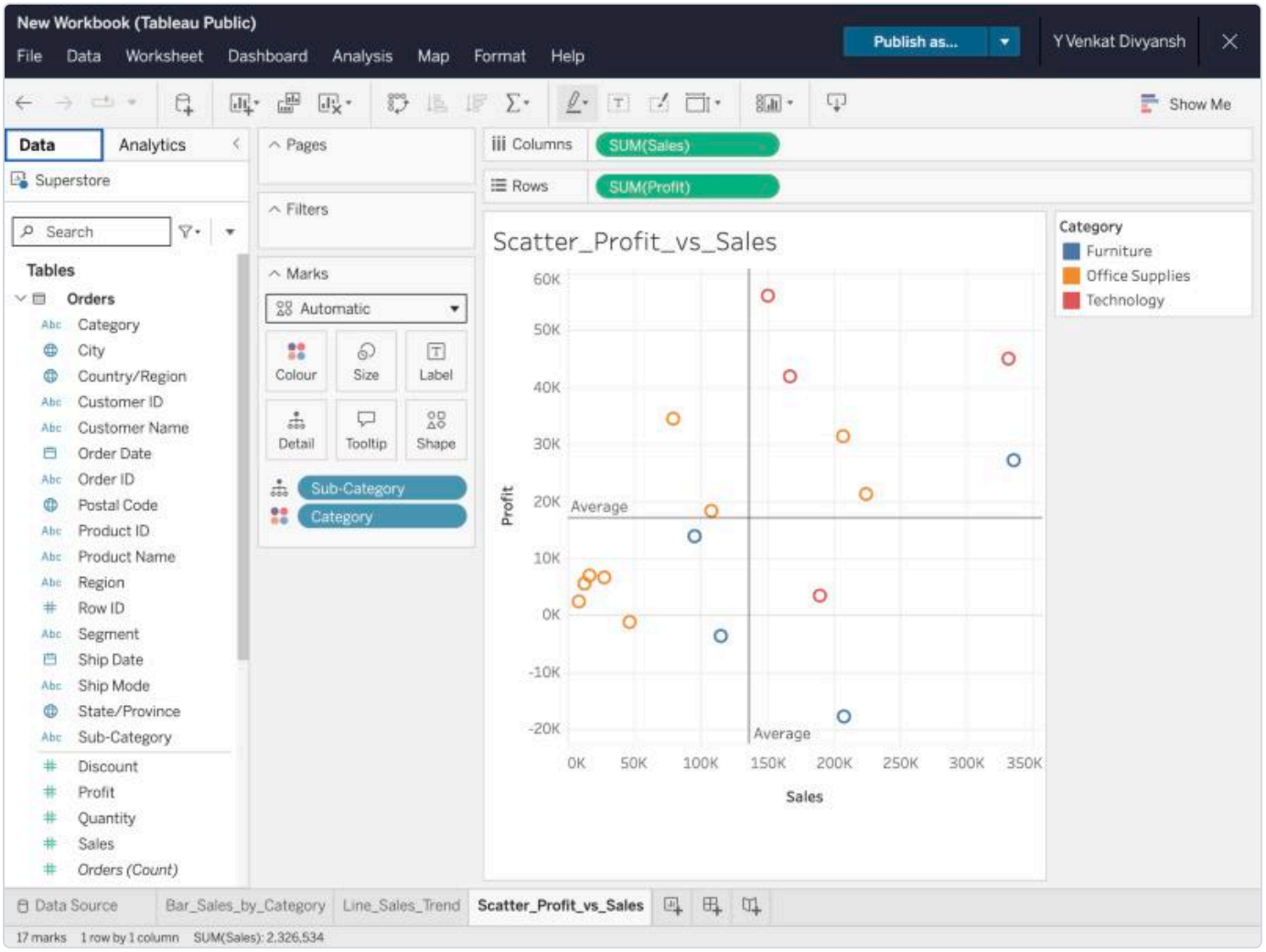


Figure 5: Worksheet 3 – Scatter_Profit_vs_Sales with average lines

> **Observation:** Technology sub-categories (red) are the most profitable; Phones has $331,843 in sales and $45,051 in profit. Some sub-categories make a loss even with high sales. The Furniture point near $207K sales has a profit of about −$17K. Points in the top-right quadrant are above average in both sales and profit.

Task 6: Build the Dashboard

1. Clicked **New Dashboard** and renamed it **Sales Dashboard**.
2. Set the size to **Fixed size → Custom, 1200 × 800**.
3. Dragged **Line_Sales_Trend** to the top, **Bar_Sales_by_Category** below it on the left, and **Scatter_Profit_vs_Sales** on the bottom right. Tableau placed them in tiled (horizontal and vertical) containers.
4. Ticked **Show dashboard title** and changed the title to **Sales and Profit Dashboard**.

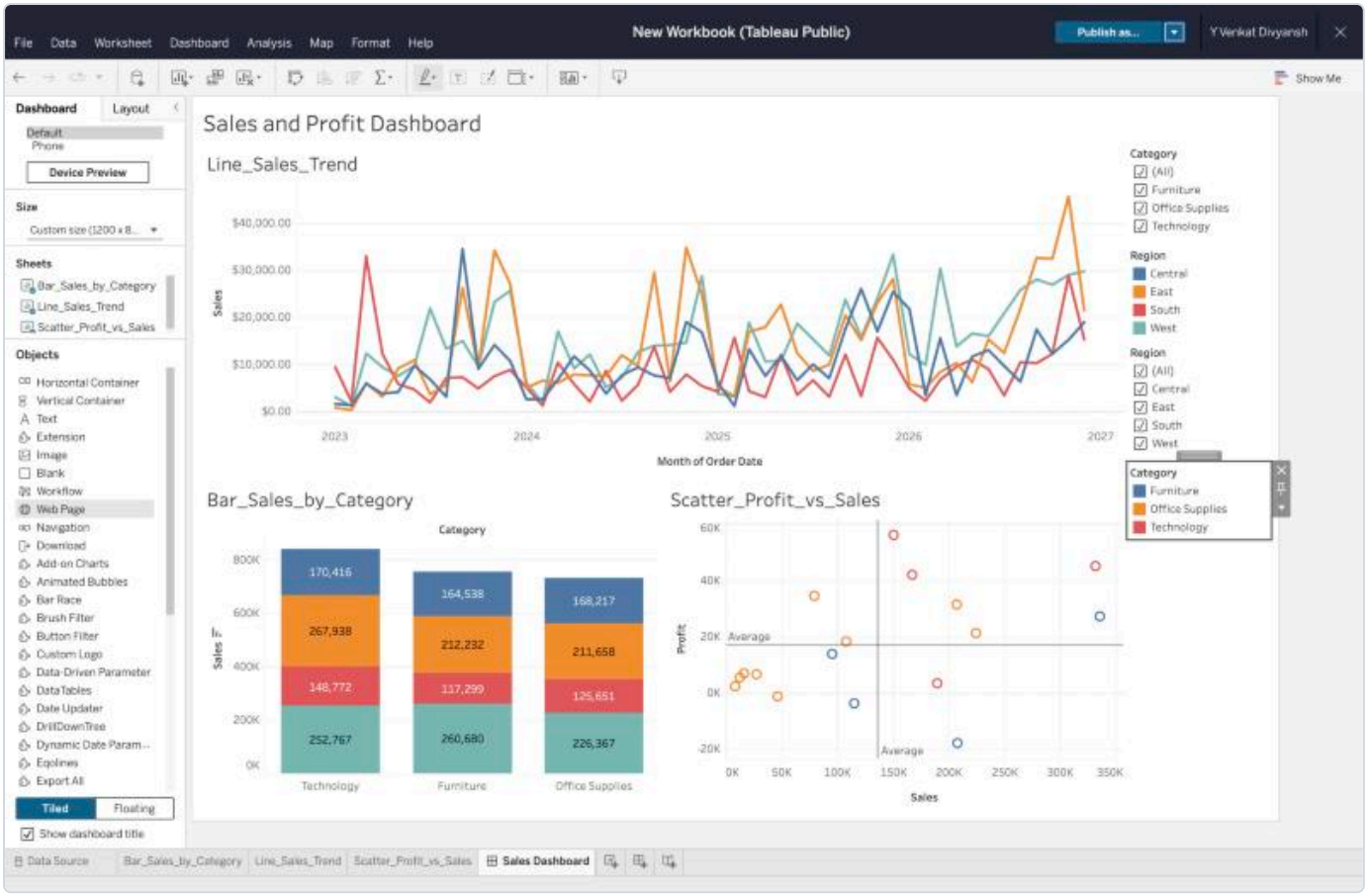


Figure 6: Sales and Profit Dashboard – line chart at the top, bar chart bottom-left, scatter plot bottom-right, with the Category and Region filters on the right

Task 7: Add Interactivity

(a) Filters

The **Category** filter (from Line_Sales_Trend) and the **Region** filter (from Bar_Sales_by_Category) were shown as filter cards on the right side of the dashboard, together with the colour legends (Figure 6).

(b) Actions

1. Opened **Dashboard → Actions → Add Action → Filter...**
2. Named the action **Filter by Category** and set *Source Sheet:* **Bar_Sales_by_Category**, *Target Sheets:* **Line_Sales_Trend** and **Scatter_Profit_vs_Sales**, *Run action on:* **Select**, and *Clearing the selection will:* **Show all values**.
3. Added a second action with **Add Action → Highlight...**, named **Highlight Category**, run on **Hover** and targeting the **Category** field, so hovering over a mark highlights the same category in the other charts.

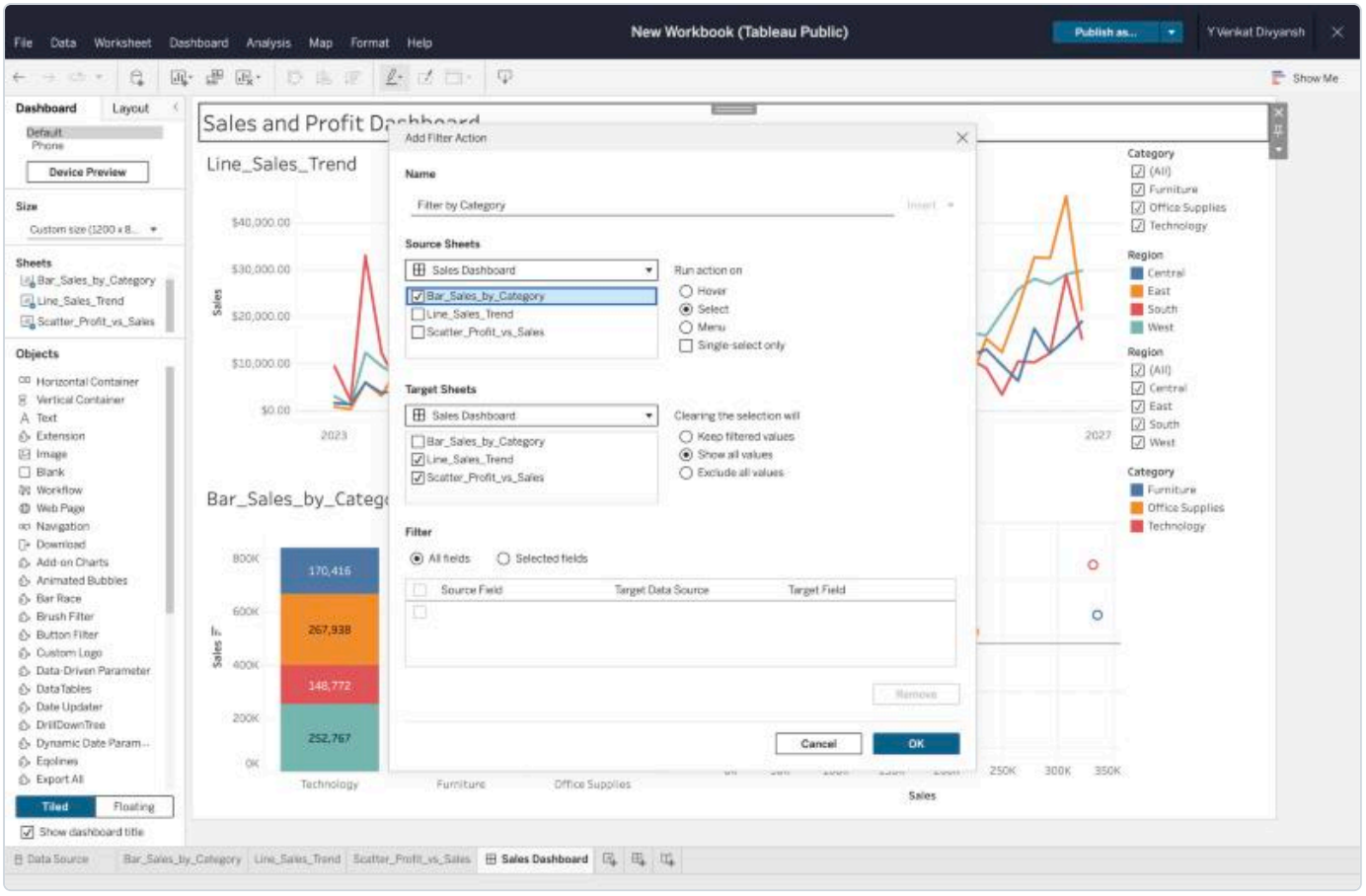


Figure 7: Add Filter Action settings

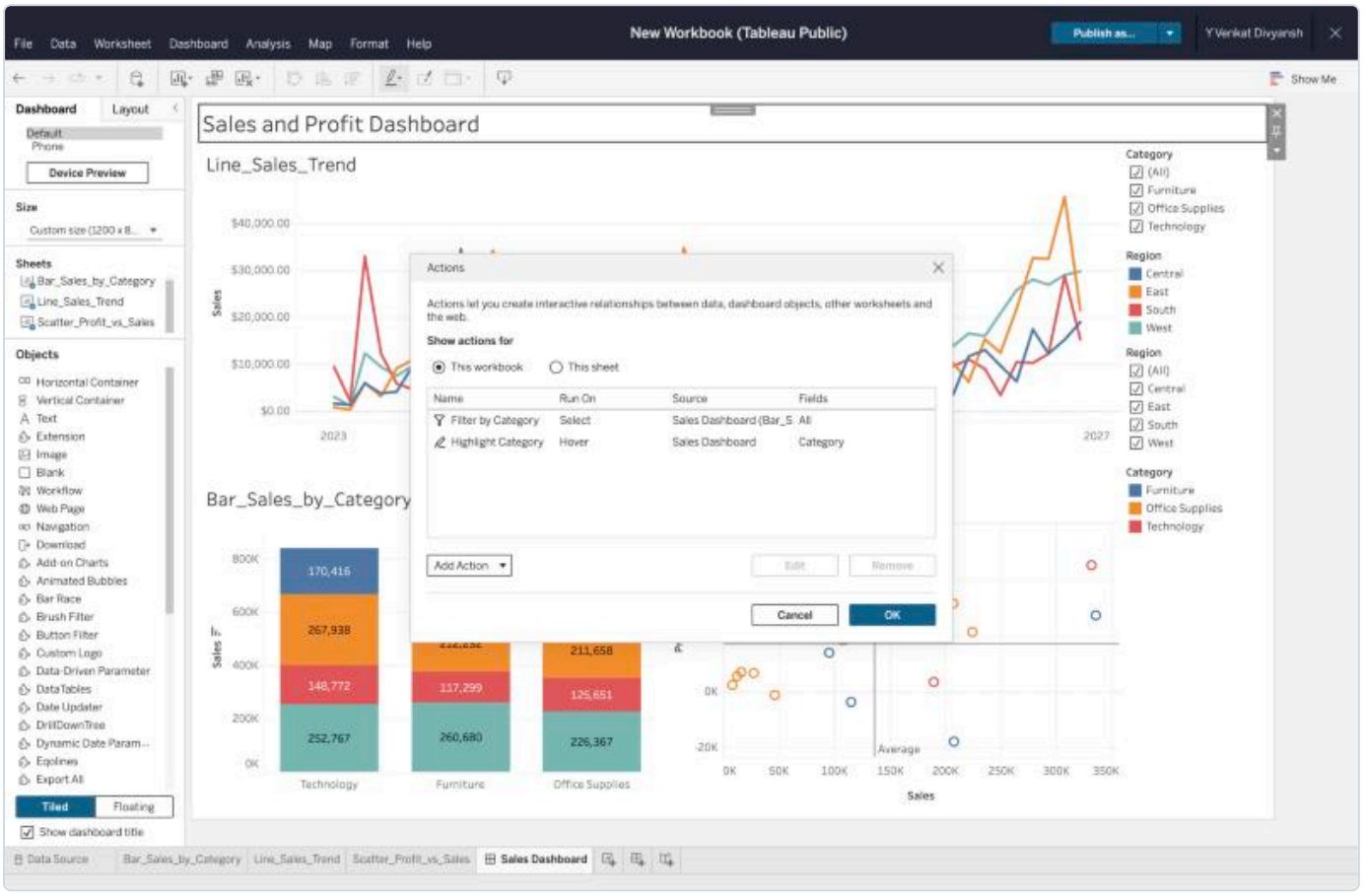


Figure 8: The two dashboard actions – Filter by Category (Select) and Highlight Category (Hover)

Testing the filter action: clicking the *Technology – East* segment of the bar chart filtered the other two charts.

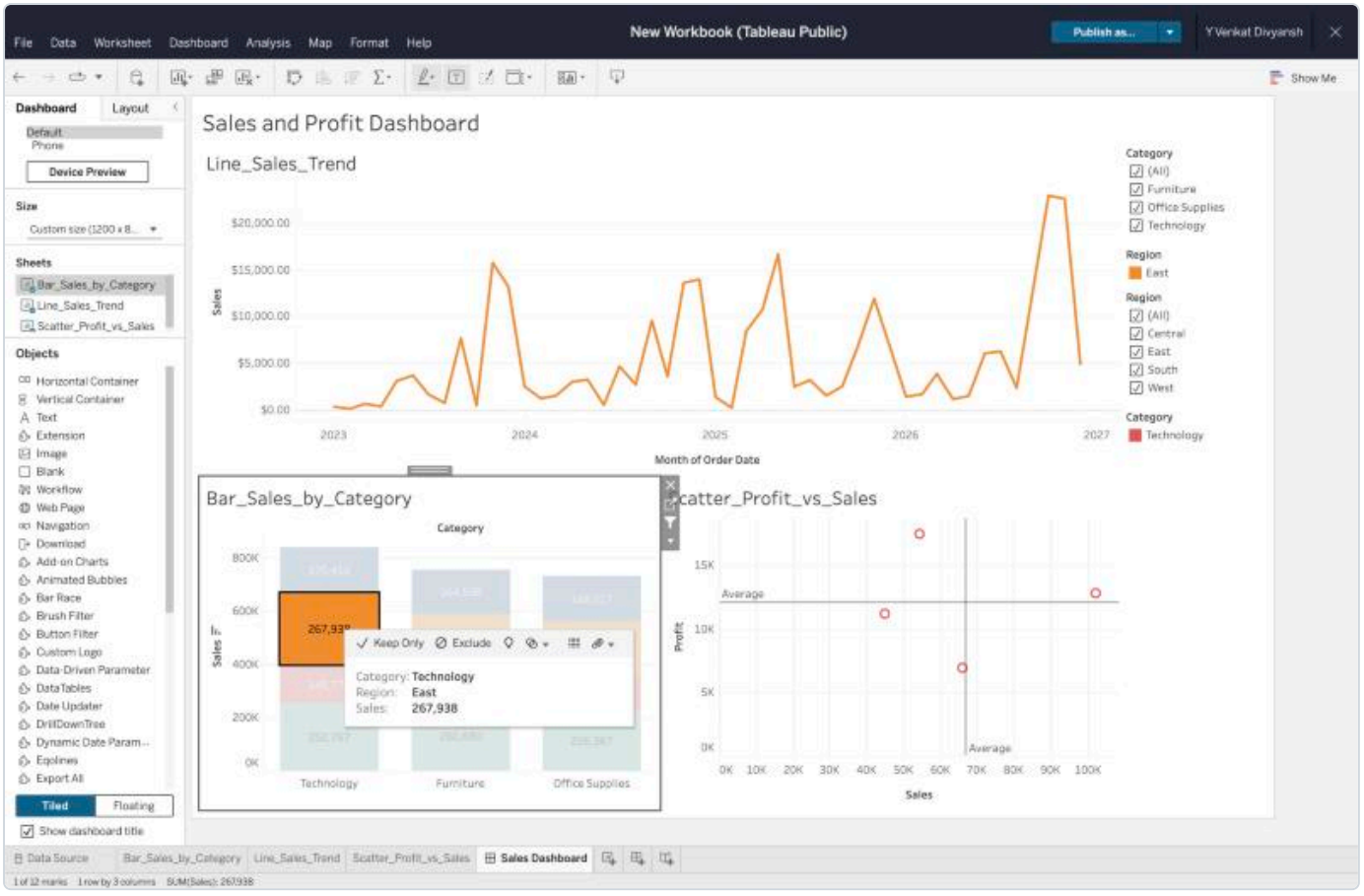


Figure 9: Filter action – selecting Technology / East in the bar chart filters the line chart and scatter plot

Testing the highlight action: hovering over the *Phones* point in the scatter plot highlighted Technology in all three charts and faded the other categories.

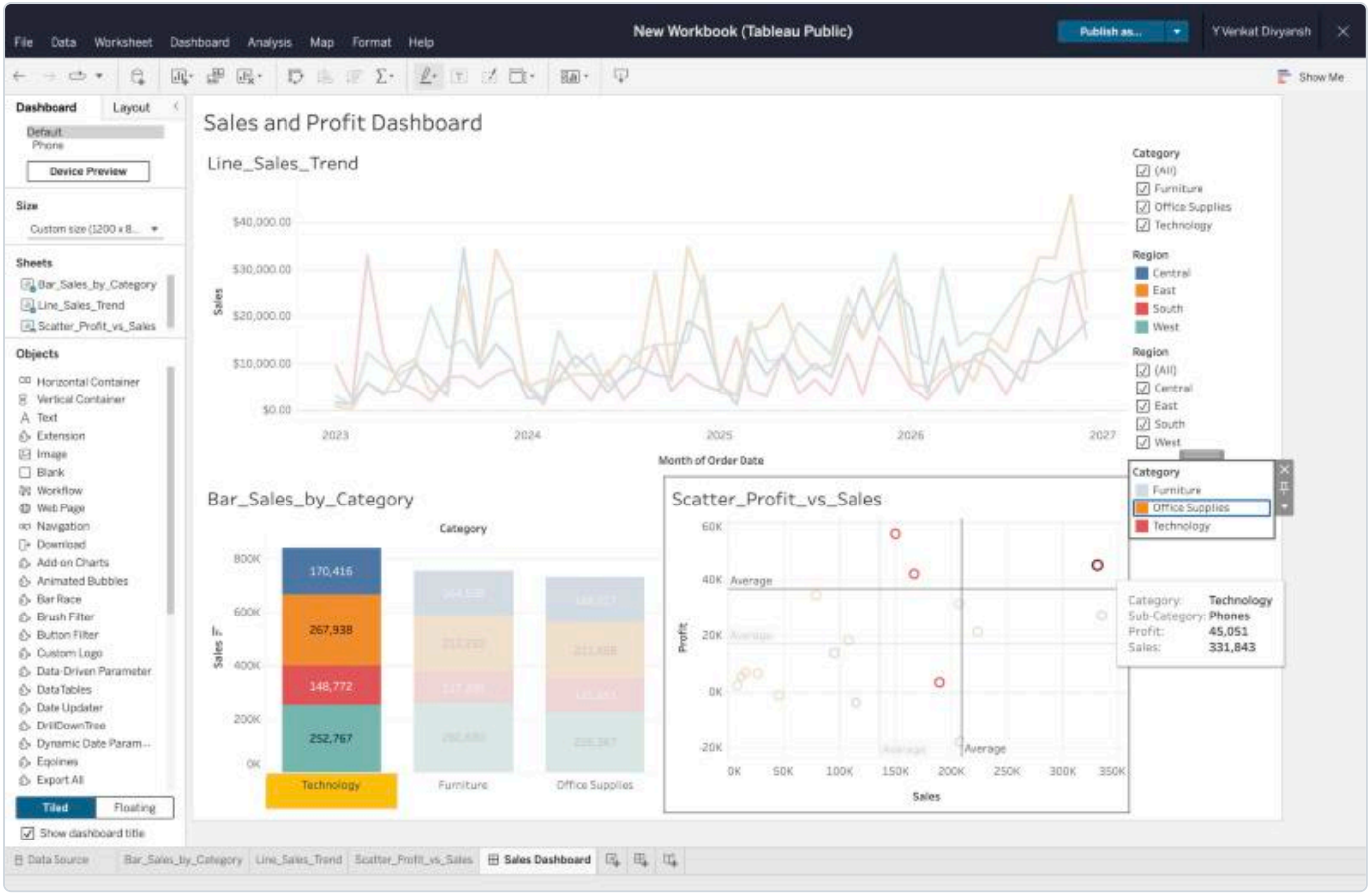


Figure 10: Highlight action – hovering over Phones highlights the Technology category across the dashboard

> **Observation:** With the filter action, selecting the Technology–East bar segment ($267,938) changed the line chart to show only Technology sales in the East, and the scatter plot to show only the four Technology sub-categories. The highlight action makes it easy to follow one category across all three charts without hiding the rest of the data.

Task 8: Publish or Export

- **To Tableau Public:** click **Publish As...** at the top right of the editor, enter a workbook name and publish. The dashboard then appears on the Tableau Public profile with a shareable link.
- **To PDF/Image:** after publishing, open the viz and use the **Download** button (Image or PDF). In Tableau Desktop, use **Dashboard → Export Image** or **File → Print to PDF**.

For this record the dashboard was **not published** to the public profile. The dashboard and every step were captured as images from Tableau Public instead (Figures 1–10).

Result

Thus an interactive Sales and Profit Dashboard was created in Tableau Public using the Sample – Superstore dataset. It contains a bar chart of sales by category coloured by region, a line chart of the monthly sales trend by region and a scatter plot of profit vs sales by sub-category with average lines. Region and Category filters, a filter action from the bar chart to the other charts, and a highlight action on Category were added and tested. The dashboard showed that Technology has the highest sales, the West and East regions sell the most, and some sub-categories make a loss despite high sales.